

RNA-Seq with Merged Data

Quality Control

To begin, I ran fastqc and multiqc on the new unprocessed data. Next, I merged the new fastq files with the previous ones, using the included `merge-fastq.sh` file. Then, I used cutadapt to remove adapter sequence.

```
sbatch ./cutadapt.sh \  
-i ./merged-fastq-exp2/fastq \  
-c ./merged-cut-fastq-exp2 \  
-a ./PolyAndIllumina.fa
```

Alignment

I aligned the samples to the complete mouse reference genome GRCm39, but used a GFF annotation file lacking rRNA records to create the gene count matrix.

```
sbatch ./Module_A/A.alignment-wrapper.sh \  
-f ./merged-cut-fastq-exp2/fastq \  
-i ./merged-cut-fastq-exp2-temp/ \  
-p ./star-params-short.txt \  
-r ./GRCm39 \  
-g ./Mus_musculus.GRCm39.114.no-rRNA.gtf \  
-a ./merged-cut-alignment-short-exp2/ \  
-q ./merged-cut-quants-short-exp2/ \  
-s ./Module_A
```

These scripts can be found on our GitHub page at https://github.com/ASU-Bioinformatics/project-scripts/tree/main/Referenced_Scripts/RNA-DEG_Modules_2025.

I also reran just the gene quantification using the complete GFF annotation file to identify the percentage of rRNA reads in each sample, so that any influence from the original rRNA amount could be accounted for in the differential expression.

Differential Analysis

I used DESeq2 in R to identify differential genes. The code for this can be found in `deseq-final.R`. Since this experiment only had two conditions, I set the design formula to compare those two directly while incorporating the rRNA percentages. Identified differential genes were filtered using a p-adjusted value <0.05 and a $\log_2$ fold change >2 .

In the principal components plot, a drift can be seen tracking with the proportion of rRNA reads in the raw data, so it was necessary to account for that in the design.

Functional Over-representation

For each direction of differential expression for each comparison, I submitted the DEG list to PANTHER and output the over-represented genes for the GO-Slim databases (more strictly curated versions of the GO ontologies). I then visualized those functional term lists as dotplots using R (this script is `pooled-dotplots.R`).